CFD interpretation practical: trimmed two-condition airfoil design

Instructions and reference material

You have **30 minutes**. Work individually. No calculator, network access, AI tool, code, or notes are permitted. Write concise, evidence-based answers in your examination booklet.

Scenario

A 625 kg cargo UAV must carry its payload at cruise and then satisfy a high-Mach dash screen. All sections are 12%-thick. The design variables are maximum camber m and camber-crest location p/c; incidence is *trimmed* separately at each flight condition to supply the required lift.

Reference relations

Use $g = 10$ N/kg and load factor $n = 1.2$:

$$W = mg, \qquad L_{\text{req}} = nW, \qquad L = q_\infty S C_L, \qquad D = q_\infty S C_D.$$

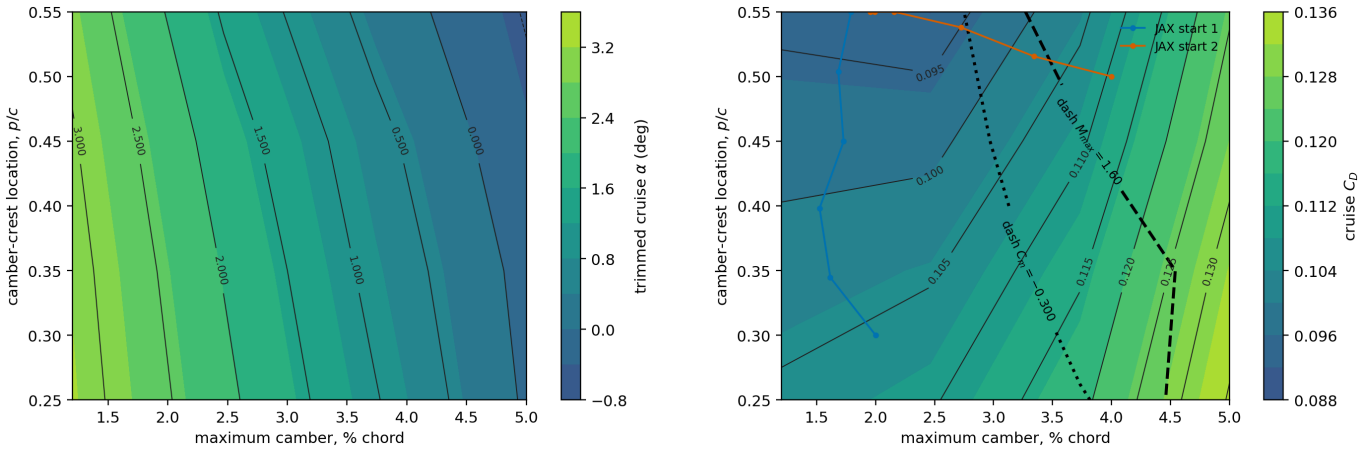
At cruise, $q_\infty S = 10.0$ kN and the trim target is $C_{L,\text{cruise}} = 0.75$. At dash, the trim target is $C_{L,\text{dash}} = 0.55$. The nominal dash limits are

$$M_{\text{max,dash}} \leq 1.60, \qquad C_{m,c/4,\text{dash}} \geq -0.300.$$

What JAX does here. At one geometry, JAX calculates how a tiny change in camber m or crest location p/c changes calculated cruise drag. The colored paths show the code taking a few steps in the lower-drag direction. JAX does *not* choose the requirements, check every limit automatically, or certify a design; those are the engineering tasks in this practical.

Model boundary. These calculations do not certify viscous drag, shock–boundary-layer interaction, separation, three-dimensional effects, structural loads, stability, or full aircraft performance.

1. Payload trim and JAX design maps [10 points]



Each map point is one airfoil after the code has adjusted incidence to meet the lift target. On the drag map: dashed black line = Mach limit; dotted black line = moment limit. The colored paths show two short JAX searches from different starting shapes.

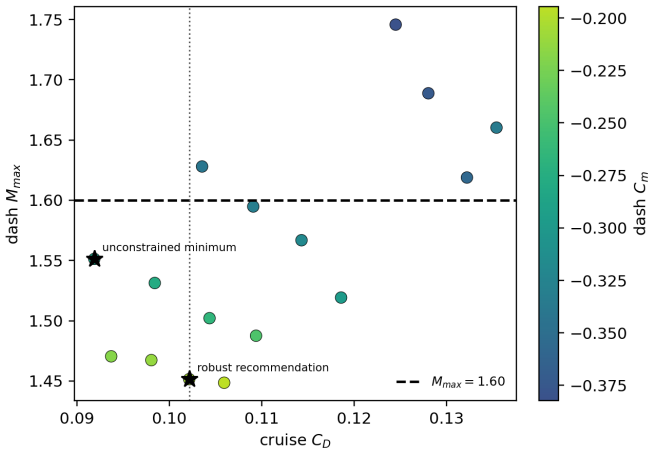
1. (2 points) Calculate L_{req} and verify $C_{L,\text{cruise}} = 0.75$ from the supplied cruise loading.

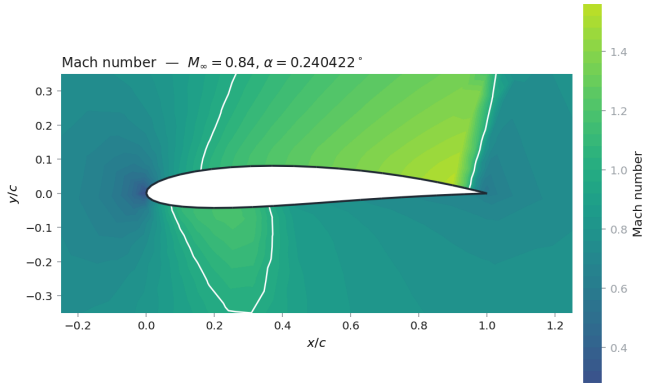
2. (2 points) The code adjusts incidence to meet lift. Why is "choose the largest camber" not a valid design rule?

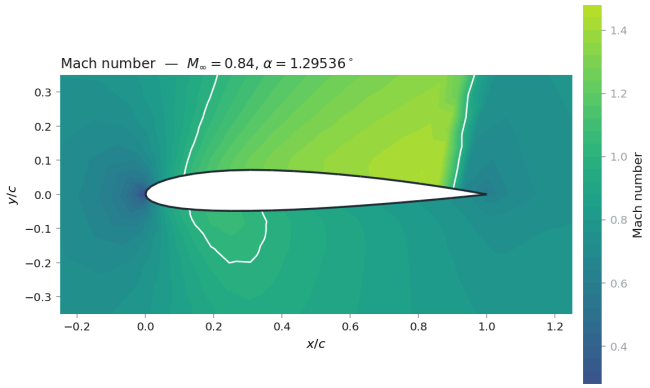
3. (3 points) On the drag map, mark the pass side of each black constraint line. Does the lowest-drag point pass both limits? Why might it still be a poor choice?

4. (3 points) In plain language, what do the colored JAX paths show? What useful information does JAX calculate before it takes the next step?

2. Robust recommendation and dash-flow evidence [10 points]







The stars show two choices. The balanced choice is not the lowest-drag point. It leaves more room below the Mach limit and above the moment limit.

Candidate	m (% chord)	p/c	$C_{D,\text{cruise}}$	$M_{\text{max,dash}}$	$C_{m,\text{dash}}$
Unconstrained minimum	2.47	0.55	0.0919	1.551	-0.288
Balanced recommendation	1.20	0.35	0.1022	1.452	-0.204

5. (3 points) Choose one candidate for the next design step. Use two numbers from the table to support your choice.

6. (3 points) Which Mach field has more margin from the dash Mach limit? Describe one visible flow difference that supports your answer.

7. (2 points) Name one next change to test. State the risk it is meant to check: shock strength, drag, moment, or separation.

8. (2 points) What numerical check is needed before trusting this drag ranking? Why is that check important near a shock?

Solution key (instructor review)

These answers correspond to the generated figure set. Award credit for concise, physically supported equivalents.

1. $W = 625(10) = 6250$ N and $L_{\text{req}} = 1.2(6250) = 7500$ N $= 7.5$ kN. Hence $C_{L,\text{cruise}} = L/(q_\infty S) = 7.5/10.0 = 0.75$.

2. Incidence is already adjusted to meet lift. More camber can change drag, dash Mach, and moment, so the largest camber is not automatically best.

3. The feasible region lies on the low-$M_{\max}$ side of the dashed contour and the allowable-moment side of the dotted contour. Here the absolute drag minimum technically passes: $M_{\max} = 1.551 < 1.60$ and $C_m = -0.288 > -0.300$. It has only 0.049 Mach and 0.012 moment margin, however, so a small numerical or model change could make it fail.

4. Each colored path starts from a different shape and moves toward lower calculated cruise drag. Before each move, JAX calculates the local drag sensitivity to camber and crest location. The student/engineer must still check the Mach and moment limits.

5. Choose the balanced recommendation. It gives up some drag (0.1022 instead of 0.0919) but has much more Mach margin (1.452 instead of 1.551) and moment margin (-0.204 instead of -0.288).

6. The balanced design has more margin: $M_{\max} = 1.452$ versus 1.551. Its Mach field has a smaller/weaker high-Mach region and a less severe upper-surface recompression. Equivalent figure-supported wording earns credit.

7. Accept a physically justified change, such as changing crest location while preserving cruise lift to control dash pressure recovery, or varying dash Mach/incidence to map shock onset. The answer must name the targeted risk: shock strength, wave drag, moment, or separation.

8. Perform a grid-refinement/mesh-convergence study, comparing trimmed incidence, C_D, $M_{\max}$, C_m, and shock location over successively finer meshes. Shock position and numerical drag are resolution-sensitive, so a small apparent margin can be numerical rather than physical.